

Lab Assignment-4

4. Programming Exercises on Classification Models and Regression Metrics

- 4.1 Write a program to apply Logistic Classification Model using Eigen, on the given data set to predict the class labels by using a threshold of 0.5. (The model is given by: $z = -4 + 1.5 \cdot x$)

x	y
1	0
2	0
3	0
4	1
5	1

- 4.2 The following table shows the actual values and predicted values obtained from a regression model. This data will be used to examine the model's performance using various regression metrics.

Actual (y)	Predicted ($\hat{y}$)
5	6
10	9
15	14
20	22
25	24

Write a program to evaluate the following regression metrics.

- (a) Mean Squared Error (MSE)
- (b) Mean Absolute Error (MAE)
- (c) R-squared